Imaging Dataset Readiness Assessment

Contoso Regional Radiology Network | Bounded AI-assisted triage workflow | 2 November 2026

OVERALL READINESS POSITION: CONDITIONAL PROCEED

Ready for configuration once Study 4, MRI-03 and CT-02 are excluded or remediated, Study 2, MRI-05 and CT-07 pass specialist review, and site-linked deviations are tagged.

19 records reviewed	11 Proceed	5 Conditional Proceed	3 Excluded or blocked

Routing triggers (Criteria, Section 4): **T1** unresolved target-region artifact; **T2** materially incomplete required series or coverage; **T3** undocumented MRI protocol deviation; **T4** scenario marks review.

Section 1 – Image Readiness Summary

> **Summary statement:** Studies 1, 3 and 5 are ready; Study 2 needs specialist review of a posterior-fossa artifact and Study 4 is blocked by an incomplete axial series, so the sample is a Conditional Proceed.

Six-dimension checklist

Study	Contrast	Orientation	Slice coverage	Artifacts	Series	Info density
Study 1 CT abdomen/pelvis	Pass	Pass	Pass	Pass	Pass	Pass
Study 2 MRI brain	Pass	Pass	Pass	Flag	Pass	Flag
Study 3 CT chest	Pass	Pass	Pass	Minor	Pass	Pass
Study 4 MRI abdomen	Pass	Pass	Fail	Pass	Fail	Flag
Study 5 CT abdomen	Pass	Pass	Pass	Pass	Pass	Pass

Minor: within acceptance. Flag: needs review. Fail: required element incomplete.

Study-level findings

Study	Evidence across six dimensions	Workflow impact	Review flag / disposition
Study 1	All six checks met.	None.	Proceed.
Study 2	Posterior-fossa artifact marked “review needed”; may affect target region.	Severity ungraded; target-region input may be degraded.	Moderate. Conditional Proceed pending specialist review (T1, T4).
Study 3	Minor noise; no major distortion.	Within acceptance.	Low. Proceed.
Study 4	Incomplete axial series; coverage gap; reduced information in affected field.	Required series incomplete; low-information input may dilute model signal.	Critical. Exclude unless slices are recovered before handoff; inclusion needs specialist coverage review (T2).
Study 5	All six checks met.	None.	Proceed.

Studies 1, 3 and 5 meet no routing trigger. Samples lack MRI-/CT- IDs and are assessed separately.

Section 2 – Modality Characteristics Documentation

> **Summary statement:** Both modalities have the structure and metadata the workflow needs, but each has a site-linked comparability issue and MRI report linkage is partial (4 of 6).

Attribute	CT documentation	MRI documentation	AI / comparability implication
Study structure	Consistent DICOM study/series; series-level acquisition/reconstruction attributes.	Sequence-level structure; sequence labels and acquisition attributes.	CT input: one reconstructed series. MRI input: a required T1/T2 set, so a missing series removes input.
Metadata	Acquisition, reconstruction, slice thickness/spacing, coverage.	Sequence type, motion, signal-quality and protocol attributes.	Supports pre-inference screening; MRI quality fields are ungraded observations.
Annotations / linked artifacts	Status recorded where applicable.	Status documented per study.	Decision criterion not provided: annotation coverage and labeling standard.
Report linkage	Complete.	Partial: 4 of 6.	Workflow dependency. Decision criterion not provided: whether triage needs reports; which two studies are unlinked.
Comparability concerns	Site A 1.25 mm; Sites B/C 2.5 mm.	Site C undocumented parameter variation.	Alters resolution and noise (CT) or contrast (MRI); site becomes a hidden input variable.
Interpretation boundaries	Data characteristics only.	Data suitability and escalation only.	Non-reference inputs are unvalidated and artifact suitability needs qualified judgment; AI outputs stay subject to radiologist review.

Connections to Section 1

- **Study 4:** sequence-level MRI structure makes the incomplete axial series blocking.
- **Study 2:** ungraded MRI quality fields explain why the artifact is routed, not excluded.
- **Studies 1, 3, 5:** Section 1 omits CT site and thickness; confirm 1.25 mm before handoff.

Section 3 – MRI Suitability Assessment

> **Summary statement:** Five of six MRI studies are suitable, MRI-05 conditionally pending sign-off on an undocumented Site C parameter change; MRI-03 is excluded for severe motion.

Sequence completeness. Required T1/T2 sequences are present in all six studies; MRI-05's unconfirmed completeness becomes a condition.

Motion and signal quality. MRI-02's minor motion is **Low**. MRI-03's severe motion "materially affects usability" (**Critical**); the record settles this, so T1 is not triggered, and motion cannot be remediated before handoff, so it is excluded.

Protocol consistency. MRI-05's undocumented Site C parameter change is a comparability issue (**Moderate**), explained by Section 2's Site C variation. Conditions: Site C documents the change and confirms no other study shares it; specialist sign-off (T3); study tagging.

Study	Sequence completeness	Motion / signal	Protocol consistency	Severity	Disposition / sign-off
MRI-01	Complete	No material motion	Reference	**None**	Proceed
MRI-02	Complete	Minor motion	Reference	**Low**	Proceed
MRI-03	Present	Severe motion	Consistent	**Critical**	Exclude; no sign-off needed
MRI-04	Complete	No material motion	Reference	**None**	Proceed
MRI-05	Present; confirm completeness	Adequate	Undocumented Site C change	**Moderate**	Conditional Proceed; sign-off required (T3)
MRI-06	Complete	No material motion	Reference	**None**	Proceed

Section 4 – CT Protocol Review

> **Summary statement:** Four CT studies match the 1.25 mm reference; CT-04/06 proceed under 2.5 mm conditions, CT-07 needs implant-artifact review, and CT-02 is blocked by an incomplete portal venous phase.

Acquisition and reconstruction. CT-04 and CT-06 use 2.5 mm, not the 1.25 mm reference: a documented deviation, not an exclusion (**Moderate**). It is systematic, reflecting the Sites B/C standard (Section 2), and affects a quarter of CT inputs.

Series completeness and coverage. CT-02's expected portal venous phase is incomplete. "Incomplete scan" is too vague: no other phase can replace this one, and the fix depends on whether it was acquired but not transferred. **Critical**.

Artifact patterns. CT-07 has severe beam hardening from a documented metallic implant; target-region overlap is unrecorded (T1): **Moderate** pending review, Critical if confirmed. The cause is patient-specific, not site- or scanner-linked.

Study	Acquisition / reconstruction	Completeness / coverage	Protocol / artifact finding	Severity	Recommendation
CT-01	1.25 mm reference	Complete	None	**None**	Proceed.
CT-02	1.25 mm where present	Portal venous phase incomplete	Missing expected phase	**Critical**	Retrieve full portal venous series from source archive, else exclude; never substitute another phase. Partial inclusion requires review (T2).
CT-03	1.25 mm reference	Complete	None	**None**	Proceed.
CT-04	2.5 mm	Complete	Reconstruction deviation	**Moderate**	Request 1.25 mm reconstruction if available; else include with thickness/site tag.
CT-05	1.25 mm reference	Complete	None	**None**	Proceed.
CT-06	2.5 mm	Complete	Reconstruction deviation	**Moderate**	As CT-04.
CT-07	1.25 mm	Complete; target represented	Severe implant beam hardening	**Moderate**	Specialist determines target-region impact (T1): include with implant tag, or exclude. Screen intake by implant field.
CT-08	1.25 mm reference	Complete	None	**None**	Proceed.

Cross-section integration check

Question	Answer	Evidence
Modality characteristic explains a Section 1 or 3 finding?	**Yes**	Study 4 (sequence structure); MRI-05 (Site C variation).
Protocol finding creates a comparability concern elsewhere?	**Yes**	CT-04/06 reflect Sites B/C standard; Site C spans both modalities.
All specialist reviews visible in the overall position?	**Yes**	Study 2, MRI-05, CT-07; Study 4 and CT-02 if included.
Decision consistent with Critical and Moderate findings?	**Yes**	Each Critical has an exclusion path; each Moderate is a condition.

Final readiness position

☐ Ready to Proceed ☒ **Conditional Proceed** ☐ Not Ready

Conditions, remediation and review requirements

1. **Exclude** MRI-03; exclude Study 4 and CT-02 unless remediated before handoff.
2. **Sign-off:** Study 2, MRI-05 and CT-07; Study 4 and CT-02 only if partial inclusion is proposed.
3. **Document** the Site C MRI change; **tag** thickness and site on every CT study.
4. **Confirm** whether triage needs report linkage and annotations; identify the two unlinked MRI studies.

Recommended next action

Send review and retrieval requests now, ahead of the one-week handoff; studies with open conditions at handoff are held out, leaving 13–18 of 19 records.